The documentation and process conversion measures necessary to comply with this revision shall be completed by 18 June 2021.

INCH-POUND

MIL-PRF-19500/557M
w/AMENDMENT 1
18 March 2021
SUPERSEDING
MIL-PRF-19500/557M
29 November 2017

PERFORMANCE SPECIFICATION SHEET

TRANSISTOR, FIELD EFFECT, N-CHANNEL, SILICON,
TYPES 2N6796, 2N6798, 2N6800, AND 2N6802
JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for a N-channel, enhancement-mode, MOSFET, power transistor intended for use in high density power switching applications. Four levels of product assurance (JAN, JANTX, JANTXV, and JANS) are provided for each encapsulated device. Two levels of product assurance are provided for each unencapsulated device.

1.2 Package outlines. The device package outlines are as follows: TO-205AF in accordance with figure 1 and leadless chip carrier (LCC) in accordance with figure 2 for all encapsulated device types. See figures 3, 4, and 5 for all unencapsulated devices.

* 1.3 Maximum ratings. Unless otherwise specified, T_A = +25°C.

Type (1)	P_T (2) T_C = +25°C	P_T T_A = +25°C	$R_{\theta JC}$ (3)	V_{DS}	V_{DG}	V_{GS}	I_{D1} (4) (5) T_C = +25°C	I_{D2} (4) T_C = +100°C	I_S	I_{DM} (6)	T_J and T_{STG}	V_{ISO} 70,000 foot altitude
	W	W	°C/W	V dc	V dc	V dc	A dc	A dc	A dc	A(pk)	°C	V dc
2N6796	25	0.8	5.0	100	100	±20	8.0	5.0	8.0	32	-55 to +150	
2N6798	25	0.8	5.0	200	200	±20	5.5	3.5	5.5	22		
2N6800	25	0.8	5.0	400	400	±20	3.0	2.0	3.0	12		400
2N6802	25	0.8	5.0	500	500	±20	2.5	1.5	2.5	10		500

(1) Electrical characteristics for "U" suffix devices are identical to the corresponding non"U" suffix devices unless otherwise specified.
(2) Derate linearly 0.2 W/°C for T_C > +25°C.
(3) See figure 6, thermal impedance curves.
(4) The following formula derives the maximum theoretical I_D limit. I_D is also limited by package and internal wires and may be limited due to pin diameter.

$$I_D = \sqrt{\frac{T_{JM} - T_C}{(R_{\theta JC}) \times (R_{DS}(on) \text{ at } T_{JM})}}$$

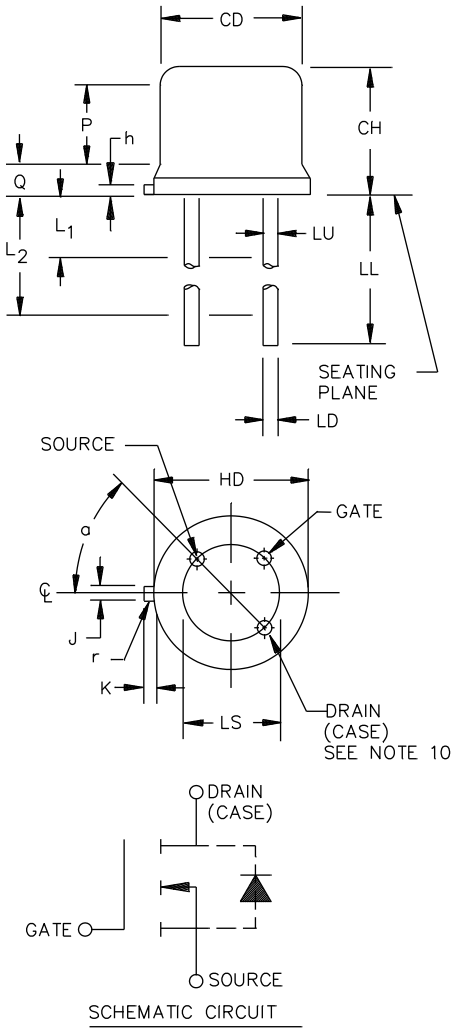
(5) See figure 7, maximum drain current graph.
(6) I_{DM} = 4 x I_{D1} as calculated in note 4.

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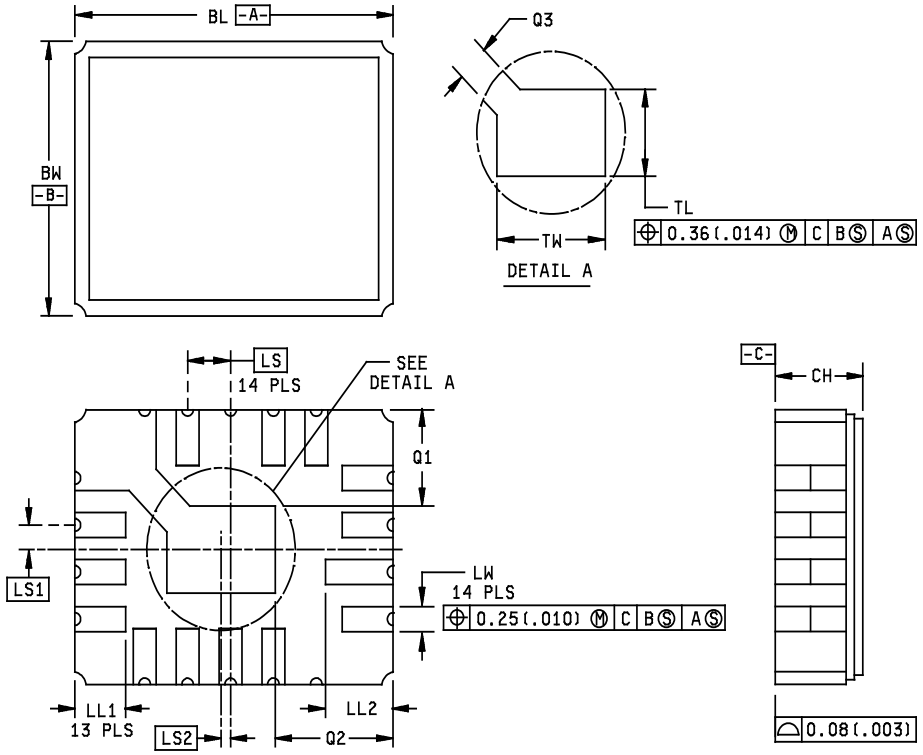


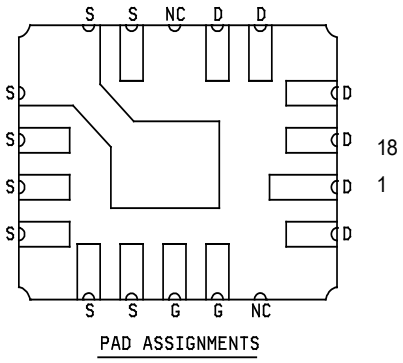
Ltr	Dimensions				
	Inches		Millimeters		Notes
	Min	Max	Min	Max	
CD	.305	.355	7.75	9.02	
CH	.160	.180	4.07	4.57	
HD	.335	.370	8.51	9.39	
h	.009	.041	0.23	1.04	
J	.028	.034	0.72	0.86	2
k	.029	.045	0.74	1.14	3
LD	.016	.021	0.41	0.53	7, 8
LL	.500	.750	12.7	19.05	7, 8
LS	.200 TP		5.08 TP		6
LU	.016	.019	0.41	0.48	7, 8
L1		.050		1.27	7, 8
L2	.250		6.35		7, 8
P	.070		1.78		5
Q		.050		1.27	4
r		.010		0.25	9
α	45° TP		45° TP		6

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Beyond radius (r) maximum, j shall be held for a minimum length of .011 (0.028 mm).
3. Dimension k measured from maximum HD.
4. Outline in this zone is not controlled.
5. Dimension CD shall not vary more than .010 (0.25 mm) in zone P. This zone is controlled for automatic handling.
6. Leads at gauge plane .054 +.001, -.000 (1.37 +0.03, -0.00 mm) below seating plane shall be within .007 (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC.
7. LU applies between L_1 and L_2. LD applies between L_2 and L minimum. Diameter is uncontrolled in L_1 and beyond LL minimum.
8. All three leads.
9. Radius (r) applies to both inside corners of tab.
10. Drain is electrically connected to the case.
* 11. In accordance with ASME Y14.5, diameters are equivalent to ϕx symbology.

FIGURE 1. Physical dimensions for TO-205AF.

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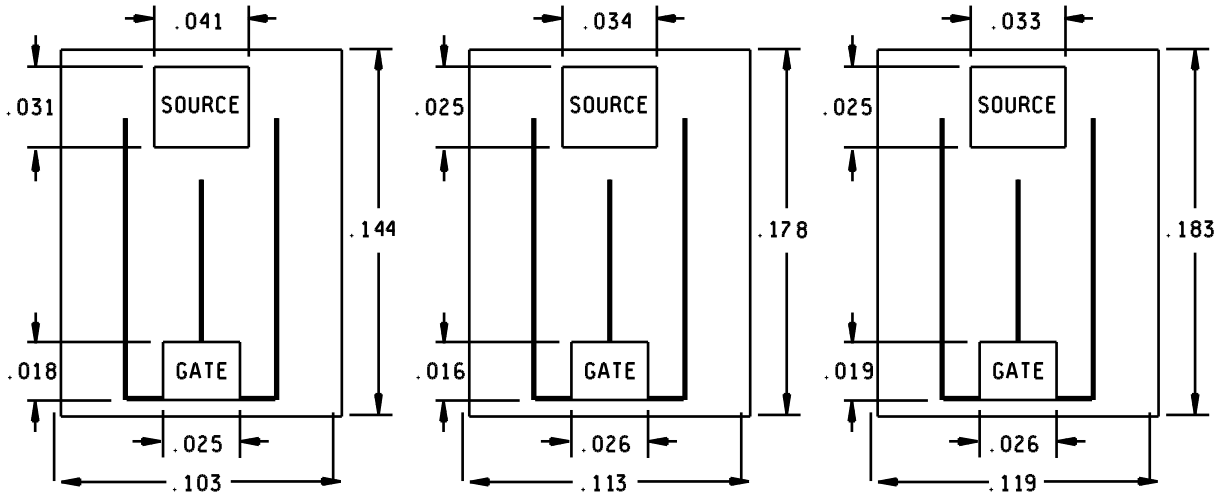


Dimensions				
Ltr	Inches		Millimeters	
	Min	Max	Min	Max
BL	.345	.360	8.77	9.14
BW	.280	.295	7.11	7.49
CH	.095	.115	2.41	2.92
LL1	.040	.055	1.02	1.40
LL2	.055	.065	1.40	1.65
LS	.050 BSC		1.27 BSC	
LS1	.025 BSC		0.635 BSC	
LS2	.008 BSC		0.203 BSC	
LW	.020	.030	0.51	0.76
Q1	.105 REF		2.67 REF	
Q2	.120 REF		3.05 REF	
Q3	.045	.055	1.14	1.40
TL	.070	.080	1.78	2.03
TW	.120	.130	3.05	3.30

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
* 3. In accordance with ASME Y14.5, diameters are equivalent to ϕx symbology.
4. Ceramic package only.

FIGURE 2. Physical dimensions for LCC.

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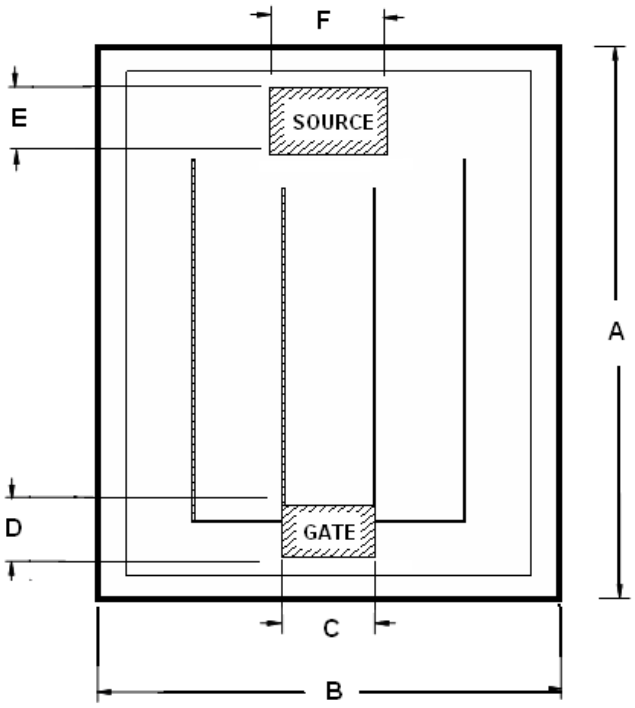
2N6796 2N6798 2N6800 and 2N6802

Inch	mm	Inch	mm	Inch	mm	Inch	mm
.016	0.41	.026	0.66	.103	2.62	.178	4.52
.018	0.46	.031	0.79	.113	2.87		
.019	0.48	.033	0.84	.119	3.02		
.0187	0.475	.034	0.86	.144	3.66		
.025	0.64	.041	1.04	.183	4.65		

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 inch (±0.13 mm).
4. The physical characteristics of the die are: the back metals are chromium, nickel, and silver. The top metal is aluminum and the back contact is the drain.
5. Die thickness is .0187 inch (0.475 mm).
* 6. Dimensions are in accordance with ASME Y14.5.

* FIGURE 3. JANHCA and JANKCA die dimensions.

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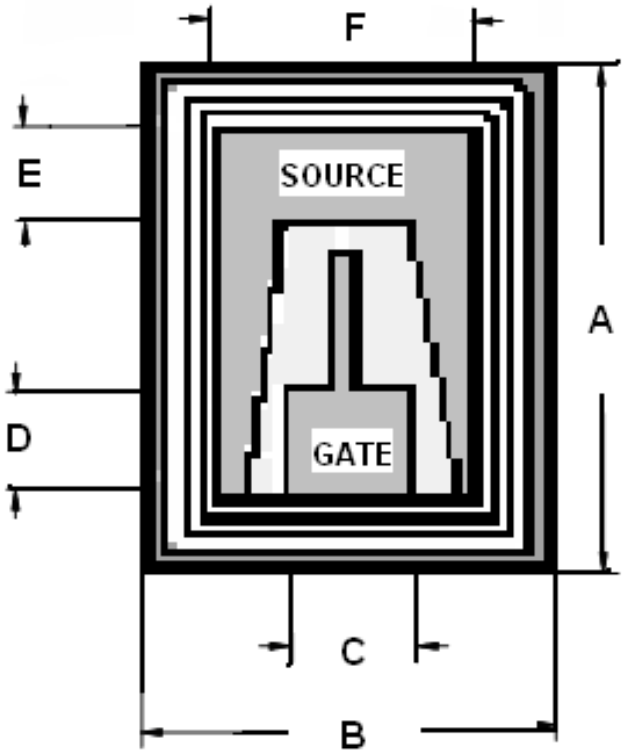


Ltr	Dimensions - 2N6796				Dimensions - 2N6798			
	Inches		Millimeters		Inches		Millimeters	
	Min	Max	Min	Max	Min	Max	Min	Max
A	.181	.185	4.60	4.70	.179	.183	4.55	4.65
B	.116	.120	2.95	3.05	.114	.118	2.89	2.99
C	.032	.034	.81	.86	.028	.030	.71	.76
D	.017	.019	.43	.48	.018	.020	.46	.51
E	.024	.026	.61	.66	.024	.026	.61	.66
F	.035	.037	.89	.94	.033	.036	.84	.91

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 inch (0.13 mm).
4. The physical characteristics of the die are: The back metals are chromium, nickel, and silver and the back contact is the drain. The top metal is aluminum.
5. Die thickness is .015 inch (0.38 mm) ±.001 inch (0.025 mm).

FIGURE 4. JANHCB and JANKCB (B-version) die dimensions for 2N6796, 2N6798.

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Ltr	Dimensions - 2N6800 and 2N6802			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.178	.182	4.52	4.62
B	.114	.118	2.89	2.99
C	.038	.040	.96	1.02
D	.038	.040	.96	1.02
E	.048	.051	1.22	1.30
F	.096	.100	2.44	2.54

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 inch (0.13 mm).
4. The physical characteristics of the die are: The back metals are Titanium, nickel, and silver and the back contact is the drain. The top metal is aluminum.
5. Die thickness is .018 inch (0.46 mm) ±.001 inch (0.025 mm).

FIGURE 5. JANHCC and JANKCC (C-version) die dimensions for 2N6800, 2N6802.

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1.4 Primary electrical characteristics at T_C = +25°C.

Type	Min $V_{(BR)DSS}$ V_{GS} = 0 V dc I_D = 1.0 mA dc	$V_{GS(th)1}$ $V_{DS} \geq V_{GS}$ I_D = 0.25 mA dc		Max I_{DSS1} V_{GS} = 0 V dc	Max $r_{DS(on)}$ (1) V_{GS} = 10 V dc	
				V_{DS} = 80 percent of rated V_{DS}	T_J = +25°C at I_{D2}	T_J = +150°C at I_{D2}
	V dc	V dc		µA dc	ohm	ohm
		Min	Max			
2N6796, U	100	2.0	4.0	25	0.18	0.39
2N6798, U	200	2.0	4.0	25	0.40	0.84
2N6800, U	400	2.0	4.0	25	1.00	2.78
2N6802, U	500	2.0	4.0	25	1.50	4.00

(1) Pulsed (see 4.5.1).

1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.5 for PIN construction example and 6.6 for a list of available PINs.

1.5.1 JAN certification mark and quality level for encapsulated devices. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: “JAN”, “JANTX”, “JANTXV” and "JANS".

1.5.2 JAN certification mark and quality level for unencapsulated devices (die). The quality level designators for unencapsulated devices (die) that are applicable for this specification sheet from the lowest to the highest level are as follows: "JANHC" and "JANKC".

1.5.3 Device type. The designation system for the device types of transistors covered by this specification sheet are as follows.

1.5.3.1 First number and first letter symbols. The transistors of this specification sheet use the first number and letter symbols "2N".

1.5.3.2 Second number symbols. The second number symbols for the transistors covered by this specification sheet are as follows: "6796", "6798", “6800”, and "6802".

1.5.3.3 Suffix letters. No suffix letters are used on devices that are packaged in the TO-205AF package of figure 1. The suffix letter "U" is used on devices that are packaged in the LCC package of figure 2.

1.5.4 Lead finish. The lead finishes applicable to this specification sheet are listed on QPDSIS-19500.

1.6 Die identifiers for unencapsulated devices (manufacturers and critical interface identifiers). The manufacturer die identifiers that are applicable for this specification sheet are "A", “B”, and "C" (see figures 3, 4, and 5).

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.
* MIL-STD-883 - Test Method Standard for Microcircuits.

* (Copies of these documents are available online at https://quicksearch.dla.mil/).

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows.

nC - nano coulomb.

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500 and on figures 1 (TO-205), 2 (LCC), 3, 4, and 5 (die) herein.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Internal construction. Multiple chip construction shall not be permitted.

* 3.4.3 Silicone die coating. The use of a silicone die coat requires a successful completion of MIL-STD-883, method 5011 on each silicone lot for its intended applications, and as part of the full MIL-PRF-19500 qualification process.

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

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* 3.6 Electrostatic discharge sensitive (ESDS). The devices covered by this specification sheet have been classified as ESDS. The devices shall be handled in accordance with the ESD program established to comply with the requirements of MIL-PRF-19500 to avoid damage due to the accumulation of static charge. The following handling practices shall be followed:

a. Devices shall be handled on benches with conductive handling devices.

b. Ground test equipment, tools, and personnel handling devices.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent, if practical.

g. Care shall be exercised, during test and troubleshooting, to apply not more than maximum rated voltage to any lead.

h. Gate must be terminated to source, $R \leq 100\ k\Omega$, whenever bias voltage is to be applied drain to source.

3.7 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.8 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table I.

3.9 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification that did not request the performance of table II tests, the tests specified in table II herein shall be performed by the first inspection lot of this revision to maintain qualification.

4.2.2 JANHC and JANKC die. Qualification shall be in accordance with MIL-PRF-19500.

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4.3 Screening (JANS, JANTX and JANTXV levels only). Screening shall be in accordance with table E-IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table E-IV of MIL-PRF-19500) (1) (2)	Measurement	
	JANS level	JANTX and JANTXV levels
(3)	Gate stress test (see 4.3.2)	Gate stress test (see 4.3.2).
(3) (4)	Unclamped inductive switching, method 3470 of MIL-STD-750 (see 4.3.3), optional	Unclamped inductive switching, method 3470 of MIL-STD-750 (see 4.3.3), optional
(3) 3c	Method 3161 of MIL-STD-750 (see 4.3.4)	Method 3161 of MIL-STD-750 (see 4.3.4)
9	Subgroup 2 of table I herein	Subgroup 2 of table I herein
10	Method 1042 of MIL-STD-750, test condition B	Method 1042 of MIL-STD-750, test condition B
11	Subgroup 2 of table I herein; ΔI_{GSSF1} = ± 20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{GSSR1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±25 μA dc or ±100 percent of initial value, whichever is greater.	Subgroup 2 of table I herein.
12	Method 1042 of MIL-STD-750, test condition A, t = 240 hours	Method 1042 of MIL-STD-750, test condition A
13	Subgroups 2 and 3 of table I herein ΔI_{GSSF1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{GSSR1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±25 μA dc or ±100 percent of initial value, whichever is greater. $\Delta r_{DS(on)1}$ = ±20 percent of initial value $\Delta V_{GS(TH)1}$= ±20 percent of initial value	Subgroup 2 of table I herein ΔI_{GSSF1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{GSSR1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±25 μA dc or ±100 percent of initial value, whichever is greater. $\Delta r_{DS(on)1}$ = ±20 percent of initial value $\Delta V_{GS(TH)1}$= ±20 percent of initial value

(1) At the end of the test program, I_{GSSF1}, I_{GSSR1}, and I_{DSS1} are measured.
(2) An out-of-family program to characterize I_{GSSF1}, I_{GSSR1}, I_{DSS1}, and $V_{GS(th)1}$ shall be invoked.
(3) Shall be performed anytime after temperature cycling, screen 3a. JANTX and JANTXV levels do not need to be repeated in screening requirements.
(4) This test is optional in screening if performed in table I, subgroup 5.

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4.3.1 Screening (JANHC and JANKC). Screening of die shall be in accordance with MIL-PRF-19500. As a minimum, die shall be 100-percent probed in accordance with table I, subgroup 2 except test current shall not exceed 20 A.

4.3.2 Gate stress test. Apply V_{GS} = ±30 V minimum for t = 250 µs minimum.

4.3.3 Unclamped inductive switching.

a. Peak current (I_D) ..... rated I_{D1}.

b. Peak gate voltage (V_{GS}) ..... 10 V dc.

c. Gate to source resistor (R_{GS}) ..... 25 Ω ≤ R_{GS} ≤ 200 Ω.

d. Initial case temperature (T_C) ..... +25°C +10°C, -5°C.

e. Inductance (L)..... 100 µH minimum.

f. Number of pulses to be applied ..... 1 pulse minimum.

g. Pulse repetition rate..... None.

* 4.3.4 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3161 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, t_{MD}, t_{SW}, (and V_H where appropriate). See table II, group E, subgroup 4 herein.

* 4.3.5 PIND. Not applicable in screening when devices are processed using alternative method and flow requirements approved by the qualifying activity, that includes incorporating the use of certified clean processing and silicone die coat. Instead, the PIND test performance shall be performed in group B3 and group C3, on a lot sample basis. PIND failures detected in group B or C will represent lot jeopardy and shall be evaluated for root cause and lot integrity.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIA (JANS) and table E-VIB (JAN, JANTX, and JANTXV) of MIL-PRF-19500 and herein.

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4.4.2.1 Quality level JANS, table E-VIA of MIL-PRF-19500.

	Subgroup	Method	Condition
	B3	1051	Test condition G.
*	B3	2052	PIND, required if not performed in screening. (22 devices, c = 0 for large lots, 12 devices, c = 0 for small lots).
	B4	1042	Test condition D. The heating cycle shall be 1 minute minimum.
*	B5	1042	Accelerated steady-state operation life; test condition A; V_{DS} = rated T_A = +175°C, t = 120 hours minimum; or, T_A = +150°C, t = 240 hours minimum. Read and record $V_{(BR)DSS}$ (pre and post) at 1 mA = I_D. Read and record I_{DSS} (pre and post). Deltas for $V_{(BR)DSS}$ shall not exceed 10 percent and I_{DSS} shall not exceed 25 μA. Accelerated steady-state gate stress; condition B, V_{GS} = rated, T_A = +175°C, t = 24 hours minimum; or, T_A = +150°C, t = 48 hours minimum.
	B5	2037	Bond strength (Al-Au die interconnects only); test condition D.

4.4.2.2 Quality levels JAN, JANTX and JANTXV, table E-VIB of MIL-PRF-19500.

Subgroup	Method	Condition
B2	1051	Test condition G.
B3	1042	Test condition D. The heating cycle shall be 1 minute minimum.
B3	2037	Test condition D. All internal bond wires for each device shall be pulled separately.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500 and as follows.

	Subgroup	Method	Condition
	C2	2036	Test condition E (not required for LCC).
*	C3	2052	PIND, required if not performed in screening. (JANS only, 22 devices, c = 0 for large lots, 12 devices, c = 0 for small lots).
	C5	3161	See 4.3.4.
	C6	1042	Test condition D. The heating cycle shall be 1 minute minimum.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Thermal impedance 2/	3161	See 4.3.4	$Z_{\theta JC}$			°C/W
Breakdown voltage, drain to source	3407	V_{GS} = 0 V dc, I_D = 1.0 mA dc, bias condition C	$V_{(BR)DSS}$			
2N6796, 2N6796U				100		V dc
2N6798, 2N6798U				200		V dc
2N6800, 2N6800U				400		V dc
2N6802, 2N6802U				500		V dc
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$ I_D = 0.25 mA dc	$V_{GS(TH)1}$	2.0	4.0	V dc
Gate current	3411	V_{GS} = +20 V dc and -20 V dc, bias condition C, V_{DS} = 0	I_{GSSF1}		± 100	nA dc
Gate current	3411	V_{GS} = +20 V dc and -20 V dc, bias condition C, V_{DS} = 0	I_{GSSR1}		± 100	nA dc
Drain current	3413	V_{GS} = 0 V dc, bias condition C, V_{DS} = 80 percent of rated V_{DS}	I_{DSS1}		25	µA dc
Static drain to source on-state resistance	3421	V_{GS} = 10 V dc, condition A, pulsed (see 4.5.1), I_D = I_{D2}	$r_{DS(on)1}$			
2N6796, 2N6796U					0.18	ohm
2N6798, 2N6798U					0.40	ohm
2N6800, 2N6800U					1.00	ohm
2N6802, 2N6802U					1.50	ohm
Static drain to source on-state resistance	3421	V_{GS} = 10 V dc, condition A, pulsed (see 4.5.1), I_D = I_{D1}	$r_{DS(on)2}$			
2N6796, 2N6796U					0.195	ohm
2N6798, 2N6798U					0.420	ohm
2N6800, 2N6800U					1.100	ohm
2N6802, 2N6802U					1.600	ohm

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - continued						
Forward voltage (source drain diode)	4011	Condition A, pulsed method, (see 4.5.1), $I_D = I_{D1}$, V_{GS} = 0 V dc	V_{SD}			
2N6796, 2N6796U					1.5	V
2N6798, 2N6798U					1.4	V
2N6800, 2N6800U					1.4	V
2N6802, 2N6802U					1.4	V
Subgroup 3						
High-temperature operation:		$T_C = T_J$ = +125°C				
Gate current	3411	V_{GS} = +20 V dc and -20 V dc, bias condition C, V_{DS} = 0	I_{GSS2}		± 200	nA dc
Drain current	3413	V_{GS} = 0 V dc, bias condition C, V_{DS} = 80 percent of rated V_{DS}	I_{DSS2}		0.25	mA dc
Gate to source voltage (thresholds)	3403	$V_{DS} \geq V_{GS}$, I_D = 0.25 mA dc	$V_{GS(TH)2}$	1.0		V dc
* Static drain to source on-state resistance	3421	Condition A, V_{GS} = 10 V dc, pulsed (see 4.5.1), $I_D = I_{D2}$	$r_{DS(on)3}$			
2N6796, 2N6796U					0.35	ohm
2N6798, 2N6798U					0.75	ohm
2N6800, 2N6800U					2.40	ohm
2N6802, 2N6802U					3.50	ohm
Low-temperature operation:		$T_C = T_J$ = -55°C				
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$, I_D = 0.25 mA dc	$V_{GS(TH)3}$		5.0	V dc
Subgroup 4						
Switching time test	3472	I_D = rated I_{D1} (see 1.3), V_{GS} = +10 V dc, R_G = 7.5Ω				
Turn-on delay time			$t_{d(on)}$			
2N6796, 2N6796U		V_{DD} = 30 V dc			30	ns
2N6798, 2N6798U		V_{DD} = 77 V dc			30	ns
2N6800, 2N6800U		V_{DD} = 176 V dc			30	ns
2N6802, 2N6802U		V_{DD} = 225 V dc			30	ns

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4 - Continued						
Rise time			t_r			
2N6796, 2N6796U		V_{DD} = 30 V dc			75	ns
2N6798, 2N6798U		V_{DD} = 77 V dc			50	ns
2N6800, 2N6800U		V_{DD} = 176 V dc			35	ns
2N6802, 2N6802U		V_{DD} = 225 V dc			30	ns
Turn-off delay time			$t_{d(off)}$			
2N6796, 2N6796U		V_{DD} = 30 V dc			40	ns
2N6798, 2N6798U		V_{DD} = 77 V dc			50	ns
2N6800, 2N6800U		V_{DD} = 176 V dc			55	ns
2N6802, 2N6802U		V_{DD} = 225 V dc			55	ns
Fall time			t_f			
2N6796, 2N6796U		V_{DD} = 30 V dc			45	ns
2N6798, 2N6798U		V_{DD} = 77 V dc			40	ns
2N6800, 2N6800U		V_{DD} = 176 V dc			35	ns
2N6802, 2N6802U		V_{DD} = 225 V dc			30	ns
Subgroup 5						
Single pulse unclamped inductive switching 3/	3470	See 4.3.3				
Safe operating area test	3474	See figure 8; t_p = 10 ms V_{DS} = 80 percent of rated V_{DS}, ($V_{DS} \leq$ 200 V dc max.)				
Electrical measurements		See table I, subgroup 2 herein.				
Subgroup 6						
Not applicable						

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 7						
Gate charge	3471	Condition B				
Test 1						
On-state gate charge			$Q_{g(on)}$			
2N6796, 2N6796U					28.51	nC
2N6798, 2N6798U					42.07	nC
2N6800, 2N6800U					34.75	nC
2N6802, 2N6802U					33.00	nC
Test 2						
Gate to source charge						
2N6796, 2N6796U			Q_{gs}		6.34	nC
2N6798, 2N6798U					5.29	nC
2N6800, 2N6800U					5.75	nC
2N6802, 2N6802U					4.46	nC
Test 3			Q_{gd}			
Gate to drain charge						
2N6796, 2N6796U					16.59	nC
2N6798, 2N6798U					28.11	nC
2N6800, 2N6800U					16.59	nC
2N6802, 2N6802U					28.11	nC
Reverse recovery time	3473	Condition A, $d_i/ d_t \leq 100$ A/µs, $V_{DD} \leq 50$ V, $I_D = I_{D1}$	t_{rr}			
2N6796, 2N6796U		$I_F = 8.0$ A			300	ns
2N6798, 2N6798U		$I_F = 5.5$ A			500	ns
2N6800, 2N6800U		$I_F = 3.0$ A			700	ns
2N6802, 2N6802U		$I_F = 2.5$ A			900	ns

1/ For sampling plan, see MIL-PRF-19500.
2/ For end-point measurements, this test is required for the following subgroups:
Group B, subgroups 2 and 3 (JANTXV).
Group B, subgroups 3 and 4 (JANS).
Group C, subgroup 2 and 6.
Group E, subgroup 1.
3/ This test is optional if performed as a 100 percent screen.

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TABLE II. Group E inspection (all quality levels) for qualification or re-qualification only.

Inspection	MIL-STD-750		Qualification and large lot quality conformance inspection
	Method	Conditions	
Subgroup 1			45 devices c = 0
Temperature cycling)	1051	Test condition G, 500 cycles	
Hermetic seal	1071		
Fine leak Gross leak			
Electrical measurements		See table I, subgroup 2	
Subgroup 2 1/			45 devices c = 0
Steady-state reverse bias	1042	Condition A, 1,000 hours	
Electrical measurements		See table I, subgroup 2	
Steady-state gate bias	1042	Condition B, 1,000 hours	
Electrical measurements		See table I, subgroup 2	
Subgroup 4			sample size N/A
Thermal impedance curves		See MIL-PRF-19500.	
Subgroup 5			
Barometric pressure	1001	2N6800, 2N6800U, 2N6802 and 2N6802U only	
Subgroup 10			
Commutating diode for safe operating area test procedure for measuring dv/dt during reverse recovery of power MOSFET transistors or insulated gate bipolar transistors	3476	Test conditions shall be derived by the manufacturer	45 devices c = 0

1/ A separate sample for each test shall be pulled.

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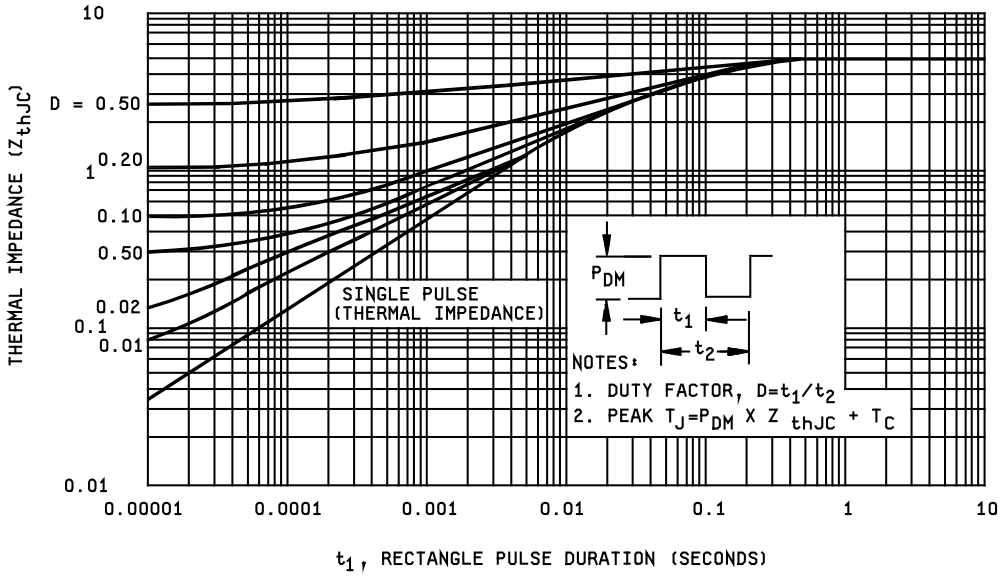
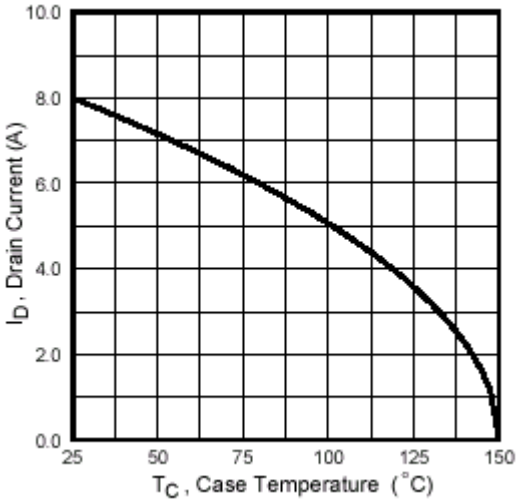
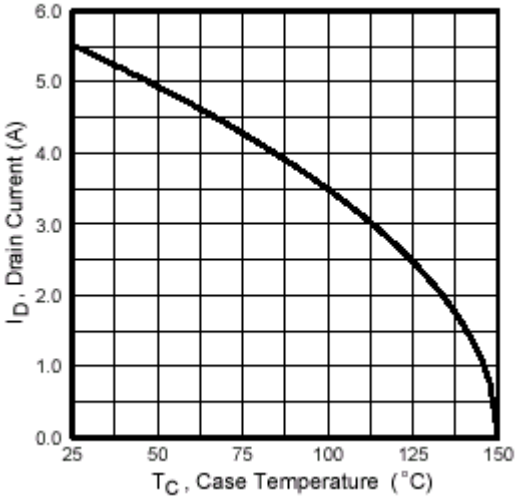


FIGURE 6. Normalized transient thermal impedance.

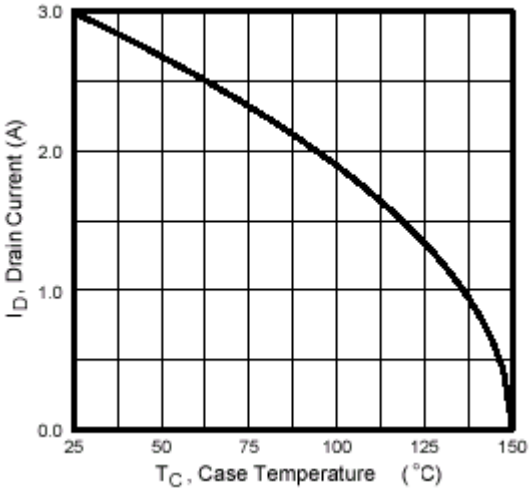
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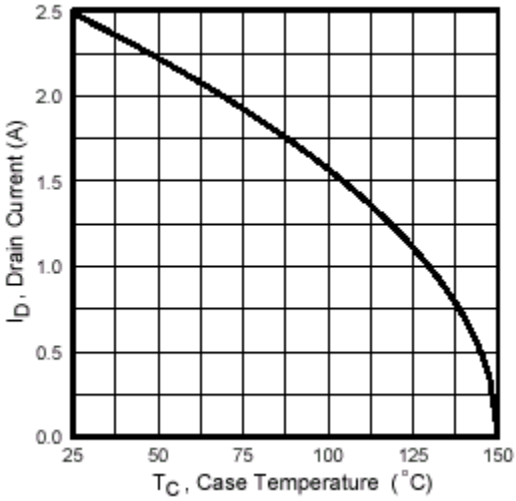
2N6796, 2N6796U



2N6798, 2N6798U



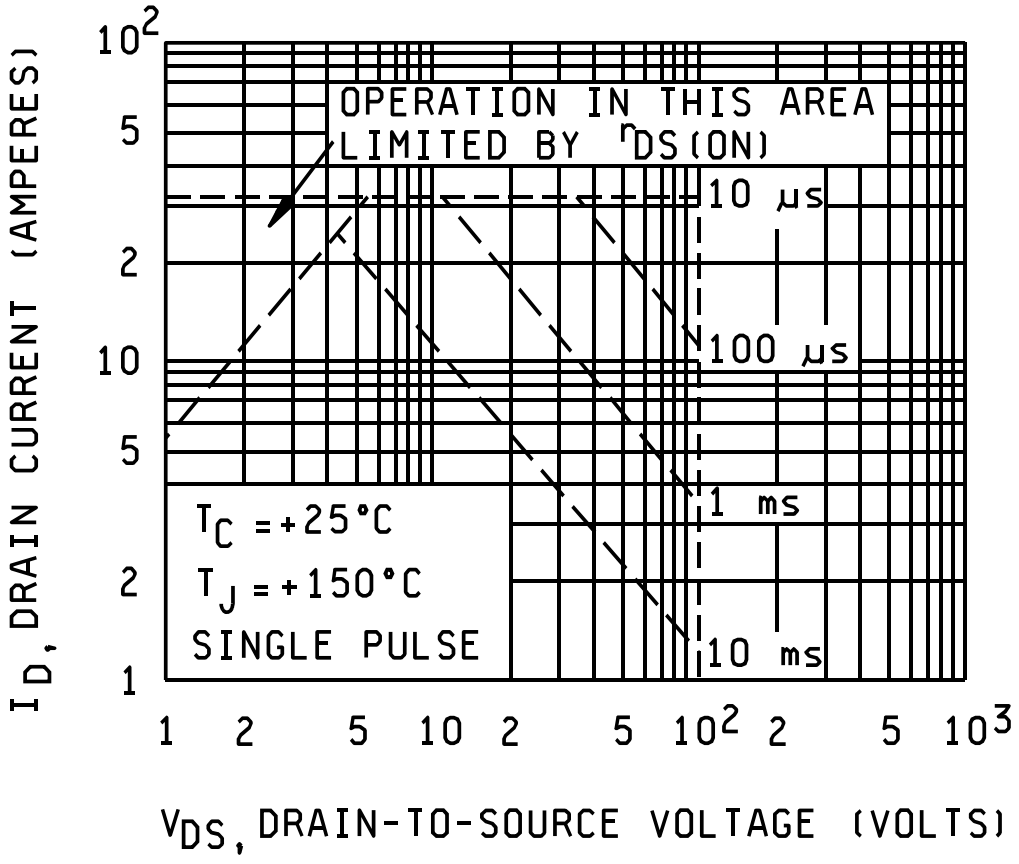
2N6800, 2N6800U



2N6802, 2N6802U

FIGURE 7. Maximum drain current versus case temperature graphs.

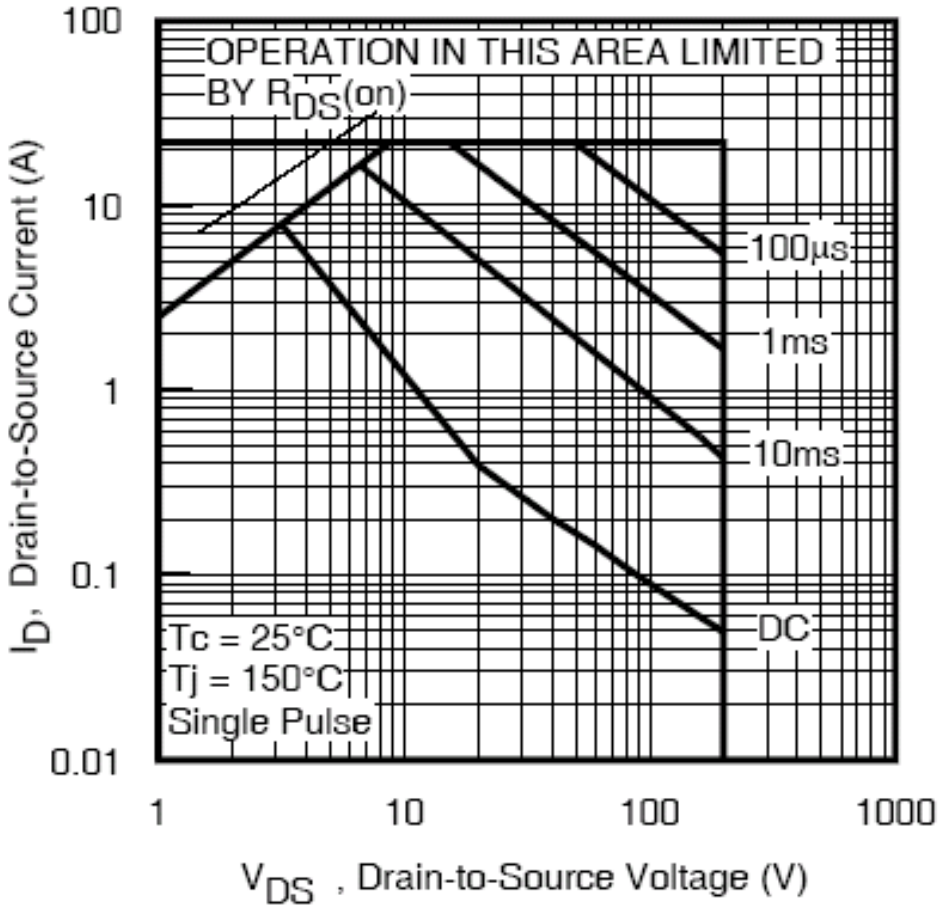
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2N6796, 2N6796U

FIGURE 8. Maximum safe operating area.

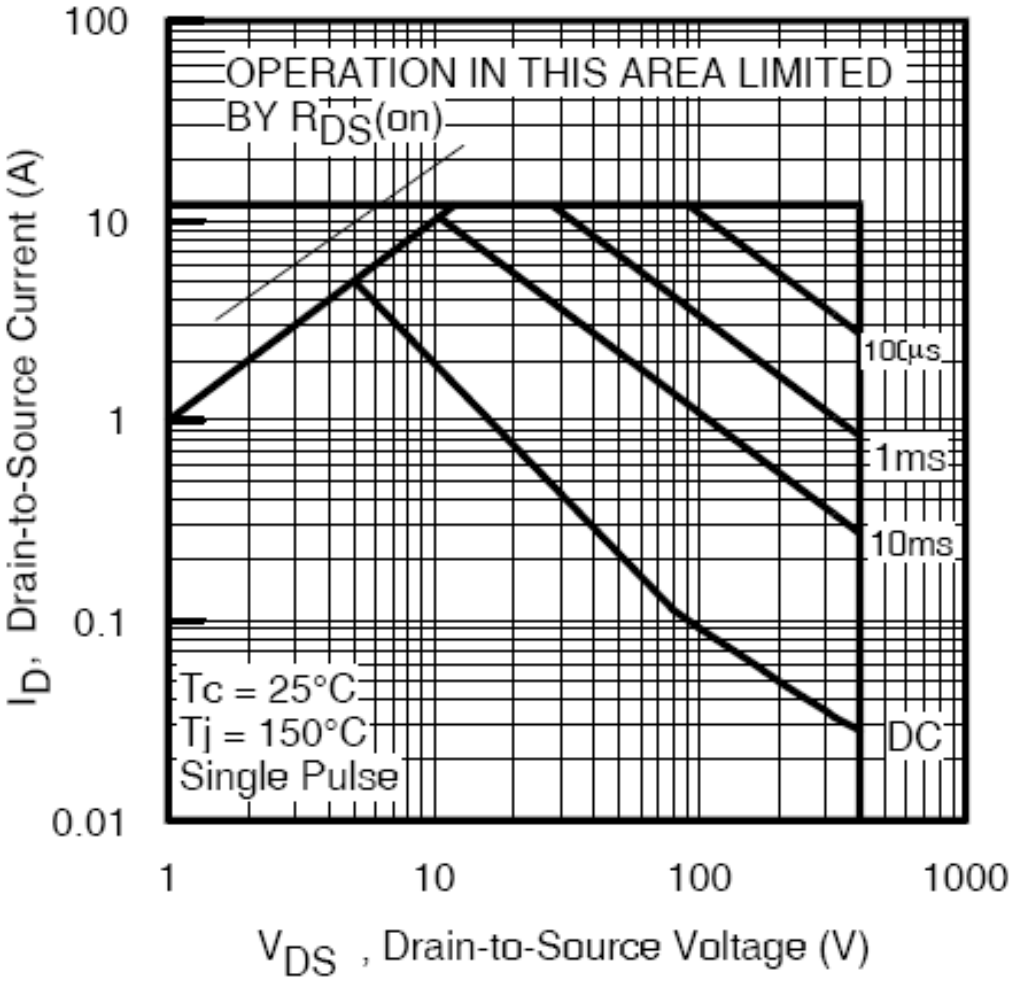
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2N6798, 2N6798U

FIGURE 8. Maximum safe operating area - Continued.

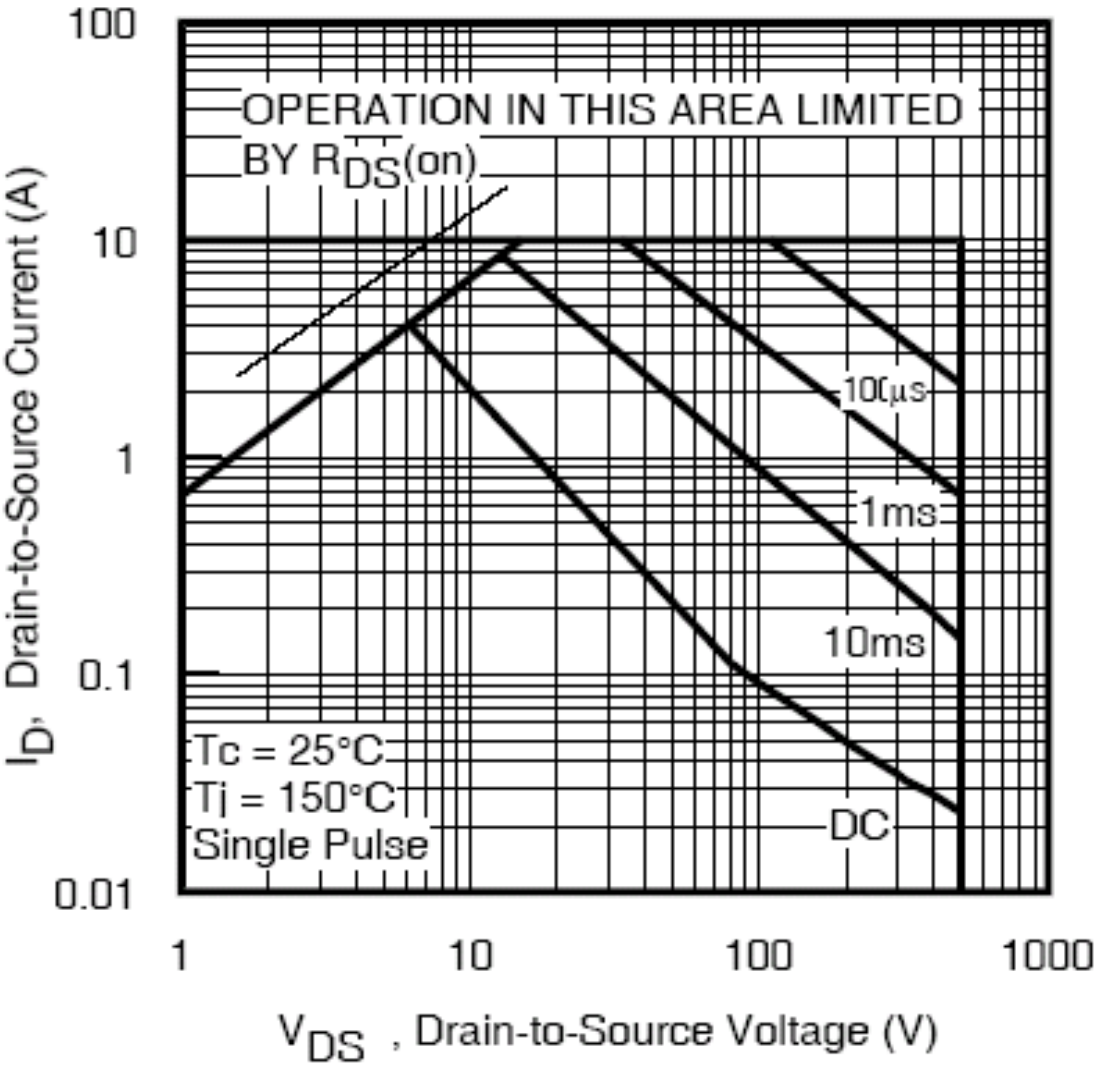
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2N6800, 2N6800U

FIGURE 8. Maximum safe operating area - Continued.

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2N6802, 2N6802U

FIGURE 8. Maximum safe operating area - Continued.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. The complete PIN, see 1.5 and 6.5.

e. For die acquisition, specify the JANHC or JANKC letter version (see figures 3, 4, and 5).

* 6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.4 Cross-reference and complement list. Parts from this specification may be used to replace the following commercial Part or Identifying Number (PIN). The term PIN is equivalent to the term part number which was previously used in this specification.

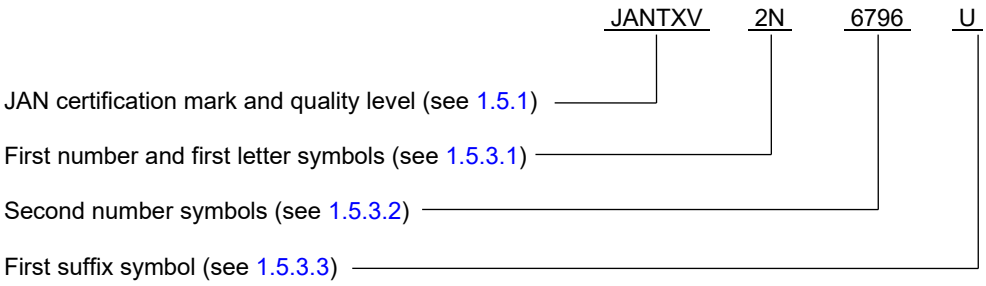
Preferred types (1)	Commercial types
2N6796	IRFF130, IRFF131, IRFF132, IRFF133
2N6798	IRFF230, IRFF231, IRFF232, IRFF233
2N6800	IRFF330, IRFF331, IRFF332, IRFF333
2N6802	IRFF430, IRFF431, IRFF432, IRFF433
2N6796U	IRFE130, IRFE131, IRFE132, IRFE133
2N6798U	IRFE230, IRFE231, IRFE232, IRFE233
2N6800U	IRFE330, IRFE331, IRFE332, IRFE333
2N6802U	IRFE430, IRFE431, IRFE432, IRFE433

(1) Prefixes are JAN, JANTX, JANTXV, or JANS.

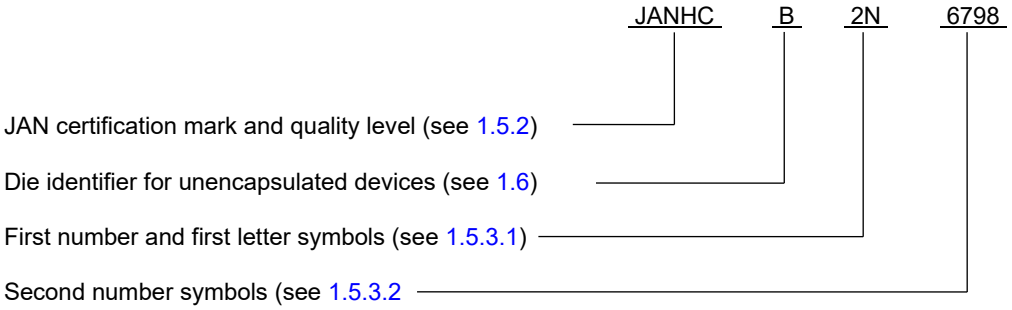
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6.5 PIN construction example.

6.5.1 Encapsulated devices The PINs for encapsulated devices are constructed using the following form.



6.5.2 Unencapsulated devices. The PINs for un-encapsulated devices are constructed using the following form.



6.6 List of PINs.

6.6.1 List of PINs for encapsulated devices. The following is a list of possible PINs for encapsulated devices available on this specification sheet.

PINs for devices of the base quality level	PINs for devices of the "TX" quality level	PINs for devices of the "TXV" quality level	PINs for devices of the "S" quality level
JAN2N6796	JANTX2N6796	JANTXV2N6796	JANS2N6796
JAN2N6796U	JANTX2N6796U	JANTXV2N6796U	JANS2N6796U
JAN2N6798	JANTX2N6798	JANTXV2N6798	JANS2N6798
JAN2N6798U	JANTX2N6798U	JANTXV2N6798U	JANS2N6798U
JAN2N6800	JANTX2N6800	JANTXV2N6800	JANS2N6800
JAN2N6800U	JANTX2N6800U	JANTXV2N6800U	JANS2N6800U
JAN2N6802	JANTX2N6802	JANTXV2N6802	JANS2N6802
JAN2N6802U	JANTX2N6802U	JANTXV2N6802U	JANS2N6802U

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6.6.2 List of PINs for unencapsulated devices. The following is a list of possible PINs for unencapsulated devices available on this specification sheet. The qualified die suppliers with the applicable letter version (example, JANHCB2N6796) will be identified on the QML.

JANC ordering information		
PIN	Manufacturers	
	69210	43611
2N6796	JANHCA2N6796 JANKCA2N6796	JANHCB2N6796 JANKCB2N6796
2N6798	JANHCA2N6798 JANKCA2N6798	JANHCB2N6798 JANKCB2N6798
2N6800	JANHCA2N6800 JANKCA2N6800	JANHCC2N6800 JANKCC2N6800
2N6802	JANHCA2N6802 JANKCA2N6802	JANHCC2N6802 JANKCC2N6802

6.7 Request for new types and configurations. Requests for new device types or configurations for inclusions in this specification sheet should be submitted to: DLA Land and Maritime, ATTN: VAC, Post Office Box 3990, Columbus, OH 43218-3990 or by electronic mail at Semiconductor@dla.mil or by facsimile (614) 692-6939 or DSN 850-6939.

6.8 Amendment notations. The margins of this specification are marked with asterisks to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
Army - CR
Navy - SH
Air Force - 85
* NASA - NA
DLA - CC

Review activities:
Army - AR, MI, SM
Navy - AS, MC
* Air Force - 19, 70, 170

Preparing activity:
DLA - CC

(Project 5961-2021-010)

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